

Hybrid CPU–GPU Based Real-Time Crowd Simulation

UCS645: Parallel and Distributed Computing
DCSE, TIET–Patiala

Problem Statement

Large-scale crowd and swarm simulations are fundamental to domains such as pedestrian dynamics, swarm robotics, and autonomous systems. These simulations require updating the state of thousands of agents at every time step while enforcing realistic local interaction rules such as separation, alignment, and cohesion.

As the number of agents increases, CPU-only implementations become computationally infeasible due to the $O(n^2)$ complexity of neighbor interaction calculations, leading to poor scalability and degraded real-time performance. This project aims to address this limitation by designing a **hybrid CPU–GPU parallel simulation framework** that efficiently distributes computation across heterogeneous processing units.

Objective

The objectives of this project are:

- To parallelize a classical agent-based simulation algorithm using GPU acceleration
- To design a hybrid execution model utilizing both CPU and GPU resources
- To evaluate scalability and performance improvements over a CPU-only baseline
- To demonstrate practical application of parallel and distributed computing concepts

Proposed Methodology

The project is based on the **Boids / Swarm AI model**, a well-known agent-based simulation approach. A hybrid parallelization strategy is adopted, where computationally intensive and data-parallel tasks are offloaded to the GPU, while control-oriented tasks are handled by the CPU.

GPU-Based Parallelization

The GPU executes the following components using CUDA kernels:

- Parallel update of agent positions and velocities
- Neighbor interaction force computation (separation, alignment, cohesion)
- Collision avoidance calculations
- Vectorized numerical computations across all agents

CPU-Based Control Logic

The CPU is responsible for:

- Managing global simulation parameters
- Enforcing zone constraints and boundary conditions
- Handling dynamic obstacles and environment updates
- Synchronizing simulation steps and visualization

Parallel and Distributed Computing Relevance

This project aligns directly with the objectives of UCS645 through:

- Parallelization of a classical algorithmic problem
- Hybrid execution on heterogeneous computing resources
- Efficient workload distribution between CPU and GPU
- Scalability and performance analysis

Performance Evaluation

The system will be evaluated using the following metrics:

- Frame time (milliseconds per simulation step)
- Frames per second (FPS)
- Scalability with increasing agent counts (5,000 to 20,000+)
- Speedup comparison between CPU-only and hybrid CPU–GPU execution

Uniqueness and Expected Outcome

The project focuses on algorithm-level parallelization rather than deep learning approaches, avoiding saturated domains such as object detection and neural network training. The expected outcome is a scalable real-time simulation capable of handling large crowds with significant performance improvements over CPU-only implementations, providing a strong demonstration of hybrid parallel computing principles.